webpdf Tutorial

The Complete Guide to HTML to PDF

Version 1.0

Frontispiece Dedication

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Chapter 1 Introduction

1.1 What is webpdf

webpdf is a lightweight HTML-to-PDF library written in C++. It renders HTML content into high-quality PDF documents without relying on a browser engine. Instead, it parses HTML and CSS directly and generates PDF instructions, resulting in fast startup times and low memory usage.

This library is particularly well-suited for:

- Server-side batch generation of PDF reports and documents
- PDF generation in embedded systems
- Applications that require precise control over PDF output
- Environments with strict startup speed and memory constraints

1.2 Key Features

Feature	Description
HTML Parsing	Supports common HTML tags and CSS styles
Chinese Support	Full support for TrueType Chinese fonts
SVG Rendering	Supports vector graphics and gradient fills
Automatic Pagination	Smart page breaks; table headers repeat across pages
Headers & Footers	Custom headers/footers with page number placeholders
Table of Contents	Auto-generated TOC with page numbers and clickable links
Full-Page Images	Use class="pagefull" for full-page rendering

> Note: webpdf is not a browser engine; it is a dedicated HTML/CSS renderer, so it only supports commonly used tags and styles. For complex web applications, consider a browser-based solution.

Chapter 2 Quick Start

2.1 Building the Project

The project uses a Makefile build system. Ensure `g++` and `make` are installed. In the project root, run:

```
cd webpdf
make
```

After a successful build, the `test_html` executable will be generated.

2.2 Running the Test

Run the test program; it reads `content.html` and outputs `output.pdf` by default:

```
./test_html
```

You can also specify input and output files:

```
./test_html input.html output.pdf
```

2.3 A Minimal Example

Here is the most basic HTML file, containing only a title and a paragraph:

```
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<title>Hello</title>
<style>
body {
    font-family: msyh;
    font-size: 24px;
```

```
}
</style>
</head>
<body>
  <h1>Hello, webpdf!</h1>
  <p>This is my first PDF document.</p>
</body>
</html>
```

> Tip: Make sure your HTML file is saved with UTF-8 encoding, otherwise Chinese characters may appear garbled.

Chapter 3 Text Styling

3.1 Fonts and Sizes

webpdf supports setting fonts and sizes via CSS. Before using a font, you must register it using the C++ API method `AddFont()`.

Example:

```
body {
  font-family: "Microsoft YaHei";
  font-size: 24px;
}
```

3.1.1 Font Weight & Italics

Use the `<b>` or `<strong>` tags for bold text, and `<i>` or `<em>` for italic text. You can also combine them: bold italic.

3.1.2 Underline and Color

Use the `<u>` tag for underlined text. Use `<font color="...">` to set text color, e.g. red, green, blue.

3.2 Text Alignment

This is left-aligned text. Left aligned text. Mixed Chinese and English display.

This is center-aligned text. Center aligned text.

This is right-aligned text. Right aligned text.

3.3 Line Height and Paragraphs

Line-height controls the vertical spacing between lines. Appropriate line-height improves readability. Generally, a value of 1.2 to 1.5 times the font size works well.

This paragraph has a small line-height (20px) with font-size 14px. The text looks rather compact, suitable for space-constrained scenarios. But too small a line-height can slow reading and cause eye fatigue. So it's not recommended for long body text.

This paragraph has a large line-height (40px) with font-size 18px. The generous spacing makes reading comfortable, especially for long articles, as readers can easily track each line. Many technical manuals and books use relaxed line-height.

Chapter 4 Tables

4.1 Basic Table

Use the `<table>` tag to create tables. Add `border="1"` to show borders.

Product Name	Price	Stock
Web Development Framework	¥299	128
Mobile App Component Library	¥499	56
Enterprise Management System	¥1999	12

4.2 Table Alignment

Cell content can be aligned both horizontally and vertically:

Left	Center	Right
Left aligned	Centered	Right aligned
Top aligned second line third line	Middle aligned	Bottom aligned

4.3 Merged Cells

Use `colspan` to merge columns and `rowspan` to merge rows.

Product Information		Price Details
Software Products	Basic Edition	¥99
	Professional Edition	¥299
	Enterprise Edition	¥999
Services	Technical Support	¥499/year
	Custom Development	Hourly rate
hardware support	Allocate servers according to projects	

4.4 Multi-Page Tables

When a table is long and spans multiple pages, add `style="position: fixed"` so that the header repeats automatically on every page for easier reading.

Feature Module	Basic	Professional
User Management	□ Supported	□ Supported

Feature Module	Basic	Professional
Role & Permissions	□ Basic	□ Full
Data Reporting	□ Basic reports	□ Advanced reports
Export	□ Excel	□ Excel/PDF/Word
Notifications	□ Not supported	□ Supported
Workflow Engine	□ Not supported	□ Supported
API Interface	□ Basic	□ Full
Third-party Integration	□ Not supported	□ Supported
Data Backup	□ Manual	□ Automatic
Operation Logs	□ Basic	□ Full audit
Multi-language	□ Not supported	□ Supported
Mobile Adaptation	□ Not supported	□ Supported
Technical Support	Community	24/7 dedicated
Upgrade Service	Minor updates	All versions
Custom Development	□ Not supported	Optional
Deployment	Public cloud	Public / Private
SLA Guarantee	99%	99.9%
Data Migration	□ Not supported	□ Supported
Training Service	Online docs	On-site training
Security Certification	Basic	Class-III (China)
Performance Tuning	Standard	Deep optimization
Plugin Marketplace	□ Not supported	□ Supported

Feature Module	Basic	Professional
Open Platform	□ Not supported	□ Supported
Multi-tenancy	□ Not supported	□ Supported
BI Analytics	□ Not supported	□ Supported
AI Assistant	□ Not supported	□ Optional
Price (per year)	¥2,999	¥9,999
Total Features	12 items	28 items

Chapter 5 Images & SVG

5.1 Raster Images

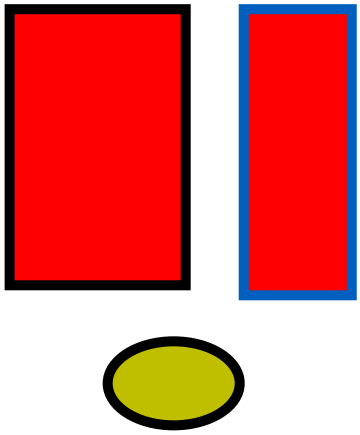
webpdf supports common raster formats like PNG and JPG. Insert images using the `<img>` tag:





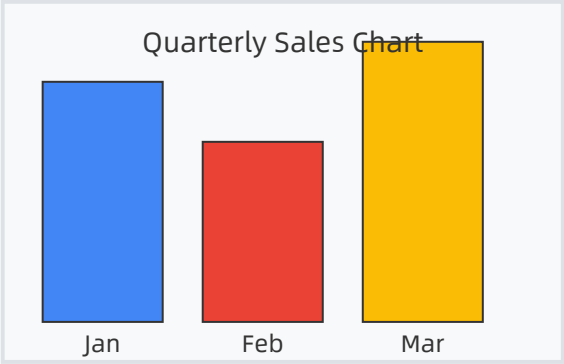
5.2 SVG Vector Graphics

SVG (Scalable Vector Graphics) scale without loss, making them ideal for charts and icons. You can reference SVG files via `<img>`, or embed SVG code directly in HTML.



5.3 Embedding SVG

Embedding SVG directly in HTML gives you finer control:

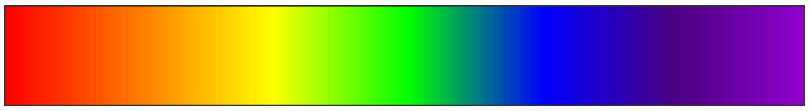


5.4 SVG Gradients

webpdf supports both linear and radial SVG gradients.

5.4.1 Linear Gradient

A linear gradient transitions colours along a straight line:

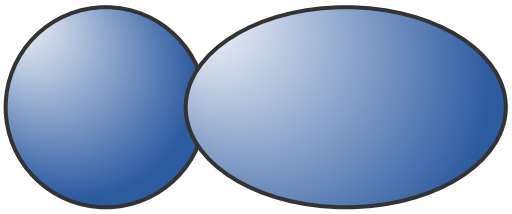


A vertical linear gradient:



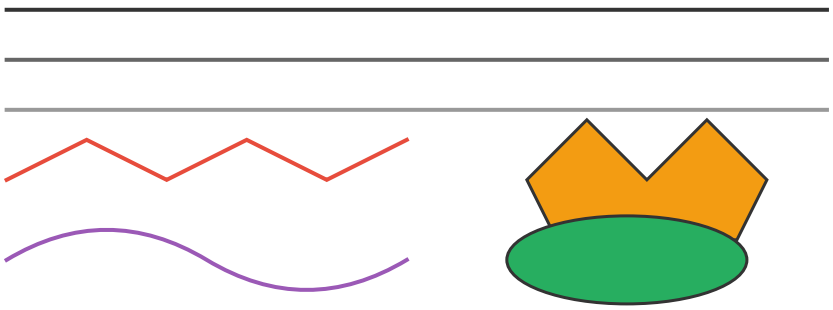
5.4.2 Radial Gradient

A radial gradient radiates from a central point, ideal for spheres and glow effects:



5.5 Paths and Shapes

SVG supports various basic shapes and path drawing:



Chapter 6 Page Layout

6.1 Forced Page Break

Sometimes you need to force a page break at a specific location. Use the `page-break-before: always` style:

```
<div style="page-break-before: always;"></div>
```

This empty div will not produce any visible content, only trigger a page break.

6.2 Full-Page Images

Use `class="pagefull"` to make an image occupy the entire page, without headers, footers, or margins. The image will be scaled and centered proportionally.

```
<p class="pagefull"><img src="images/fullpage.png"></p>
```

Here is a full-page PNG image:



The image above fills an entire page, with no header or footer. Now we are on a new page, headers and footers are back to normal.

SVG images can also be full-page:



This is another new page, with headers and footers visible. The full-page feature is perfect for covers, illustrations, charts, or any content that deserves a whole page.

> Pro tip: The `pagefull` class can be applied to any container; the first `<img>` inside will be used for full-page rendering, while the container itself and any other children will not be displayed.

6.3 Containers and Backgrounds

Use `<div>` to create block containers with background colours, borders, rounded corners, etc.:

Info Box

This is a blue-themed info box. Use it for important messages, notes, etc.

Warning

This is a warning box with a prominent red left border.

HOT

Hot Recommendation

A container with a background image, rounded border, and padding.

Fieldset Info Box

This is another info box using the fieldset element for a grouped style.

Chapter 7 Advanced Features

7.1 Hyperlinks

Use the `<a>` tag to create hyperlinks; in the PDF, clicking them will open the URL. For example, visit Example Website for more info.

Long links wrap correctly: https://www.example.com/very/long/path/to/document.html?query=value&param=test&foo=bar&baz=qux

7.2 Horizontal Rules

Use the `<hr>` tag to insert horizontal rules for separating content:

Default style:

Red thick rule:

Green dashed rule:

Blue dotted rule, 50% width centered:

Orange short rule, right-aligned:

7.3 Headers and Footers

Use `class="pageheader"` and `class="pagefooter"` to define custom headers and footers for every page. This document uses them.

Two special placeholders are available in headers/footers:

Placeholder	Description
{pagenum}	Current page number

{pagetotal}	Total page count

> Note: Cover pages and TOC pages do not show headers/footers by default. Full-page (`pagefull`) pages also hide them.

7.4 Table of Contents Generation

webpdf can automatically generate a table of contents. Add `class="catalogue1"`, `catalogue2`, or `catalogue3` to headings to include them in the TOC.

Class Name	TOC Level	Corresponding Headings
catalogue1	Level 1	h1 ~ h3
catalogue2	Level 2	h4 ~ h5
catalogue3	Level 3	h5 ~ h6

The generated TOC includes page numbers and clickable links to each section.

7.5 Cover and Title Pages

Use `class="pagecover"` to create a cover page. Cover pages have no headers/footers and content starts from the top of the page.

Use `class="pagecatalogue"` for a TOC background, combined with `class="cataloguecontent"` to auto-insert the TOC.

Chapter 8 API Reference

8.1 The FPDF Class

`FPDF` is the core PDF document class, used to create and manipulate PDF files.

Method	Description
`FPDF()`	Constructor – creates a new PDF document
`AddPage()`	Adds a new page
`AddFont(family, style, file, dir)`	Registers a font file
`SetFont(family, style, size)`	Sets the current font
`WriteHTML(html, line_height)`	Renders HTML content into the PDF
`Output(dest, name)`	Outputs the PDF file
`SetShowHeader(show)`	Enables/disables header display
`SetShowFooter(show)`	Enables/disables footer display
`SetShowHeaderFooter(show)`	Sets both header and footer visibility
`PageNo()`	Returns the current page number
`SetPageNumberOffset(offset)`	Sets a page number offset
`SetLeftMargin(margin)`	Sets left margin
`SetRightMargin(margin)`	Sets right margin
`SetTopMargin(margin)`	Sets top margin
`SetAutoPageBreak(auto, margin)`	Configures automatic page breaking
`GetPageWidth()`	Returns page width
`GetPageHeight()`	Returns page height
`GetX() / GetY()`	Returns current X/Y coordinates

Method	Description
`SetXY(x, y)`	Sets current position
`Image(file, x, y, w, h)`	Inserts an image
`Rect(x, y, w, h, style)`	Draws a rectangle
`SetFillColor(r, g, b)`	Sets fill colour
`SetDrawColor(r, g, b)`	Sets stroke colour
`SetTextColor(r, g, b)`	Sets text colour
`SetLineWidth(width)`	Sets line width
`AddAxialShading(...)`	Adds an axial (linear) gradient
`ShadeFill(x, y, w, h, id)`	Fills a rectangle with a gradient

8.2 Complete Example Code

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
#include "fpdf.h"

std::string read_file(const std::string& path) {
    std::ifstream f(path);
    if (!f.is_open())
        throw std::runtime_error("Cannot open file: " + path);
    std::stringstream ss;
    ss << f.rdbuf();
    return ss.str();
}

int main(int argc, char* argv[]) {
    try {
        std::string html_file = "content.html";
        std::string pdf_file = "output.pdf";

        if (argc > 1) html_file = argv[1];
        if (argc > 2) pdf_file = argv[2];
```

```
        std::string html = read_file(html_file);

        webpdf::FPDF pdf;
        pdf.AddPage();
        pdf.AddFont("msyh", "", "msyh.ttf", "font/");
        pdf.SetFont("msyh", "", 14);
        pdf.WriteHTML(html);
        pdf.Output("F", pdf_file);

        std::cout << "PDF generated: " << pdf_file << std::endl;
        return 0;
    } catch (const std::exception& e) {
        std::cerr << "Error: " << e.what() << std::endl;
        return 1;
    }
}
```

Chapter 9 Frequently Asked Questions

9.1 Text Issues

Problem	Solution
Chinese characters appear as boxes or garbled	Ensure you have registered a Chinese font with AddFont() and that the font-family in CSS matches exactly
Font weight (bold) does not work	You must register a separate font file for that weight; webpdf does not simulate bold automatically
Line wrapping is incorrect	Check for special characters or spaces; Chinese wraps based on character width

9.2 Image Issues

Problem	Solution
Images do not appear	Check that the image path is correct and relative to the program's working directory

SVG renders as black	Verify SVG syntax; gradients must be defined inside <defs>
SVG gradients show incorrectly	Ensure the gradient ID is correct and referenced as url(#id)

9.3 Pagination Issues

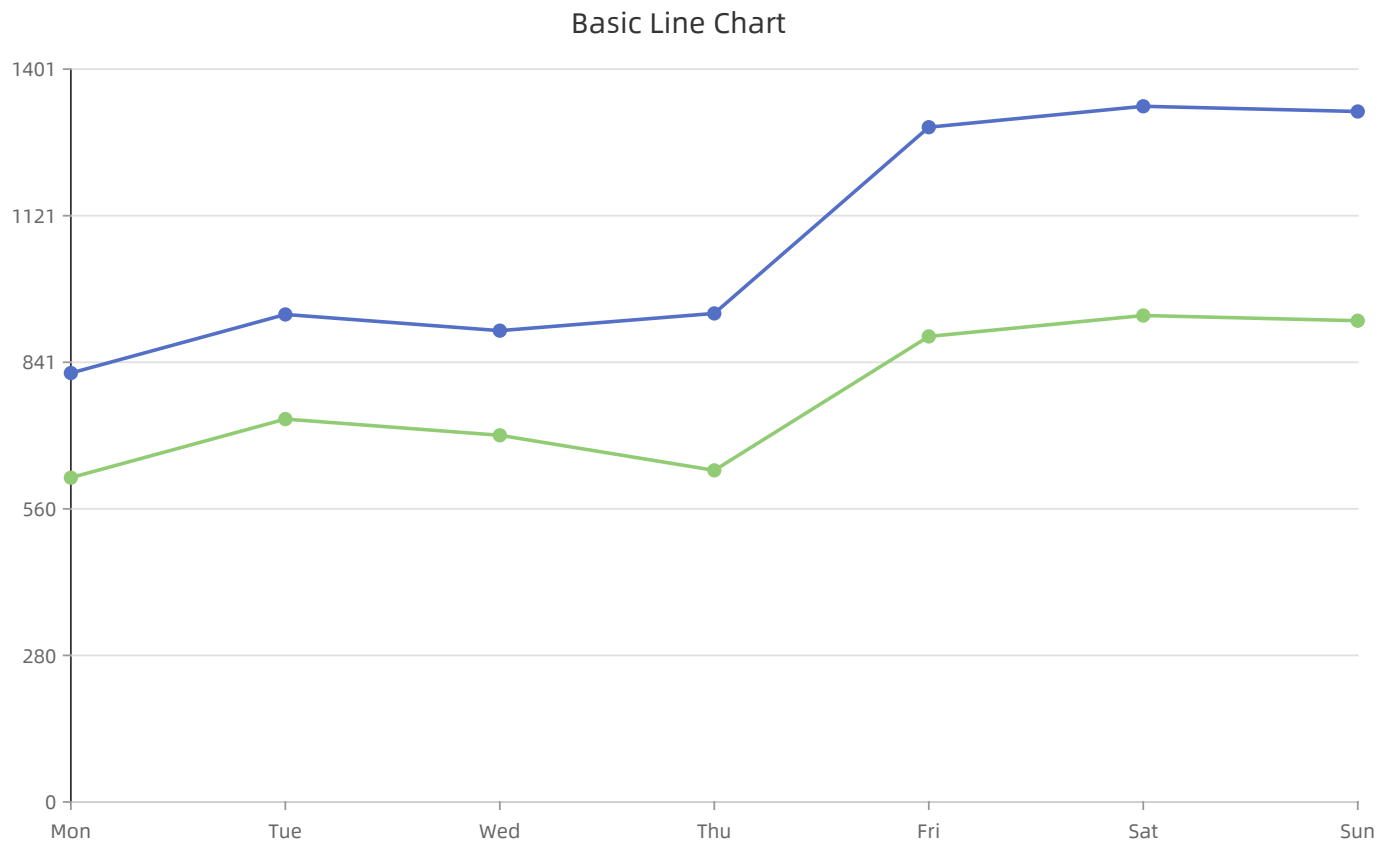
Problem	Solution
Blank page before a forced page break	Use an empty without any content inside
Table header does not repeat across pages	Add style="position: fixed" to the <table> tag
Full-page image still shows headers/footers	Make sure class="pagefull" is on the container and it contains an <img> child

9.4 Other Issues

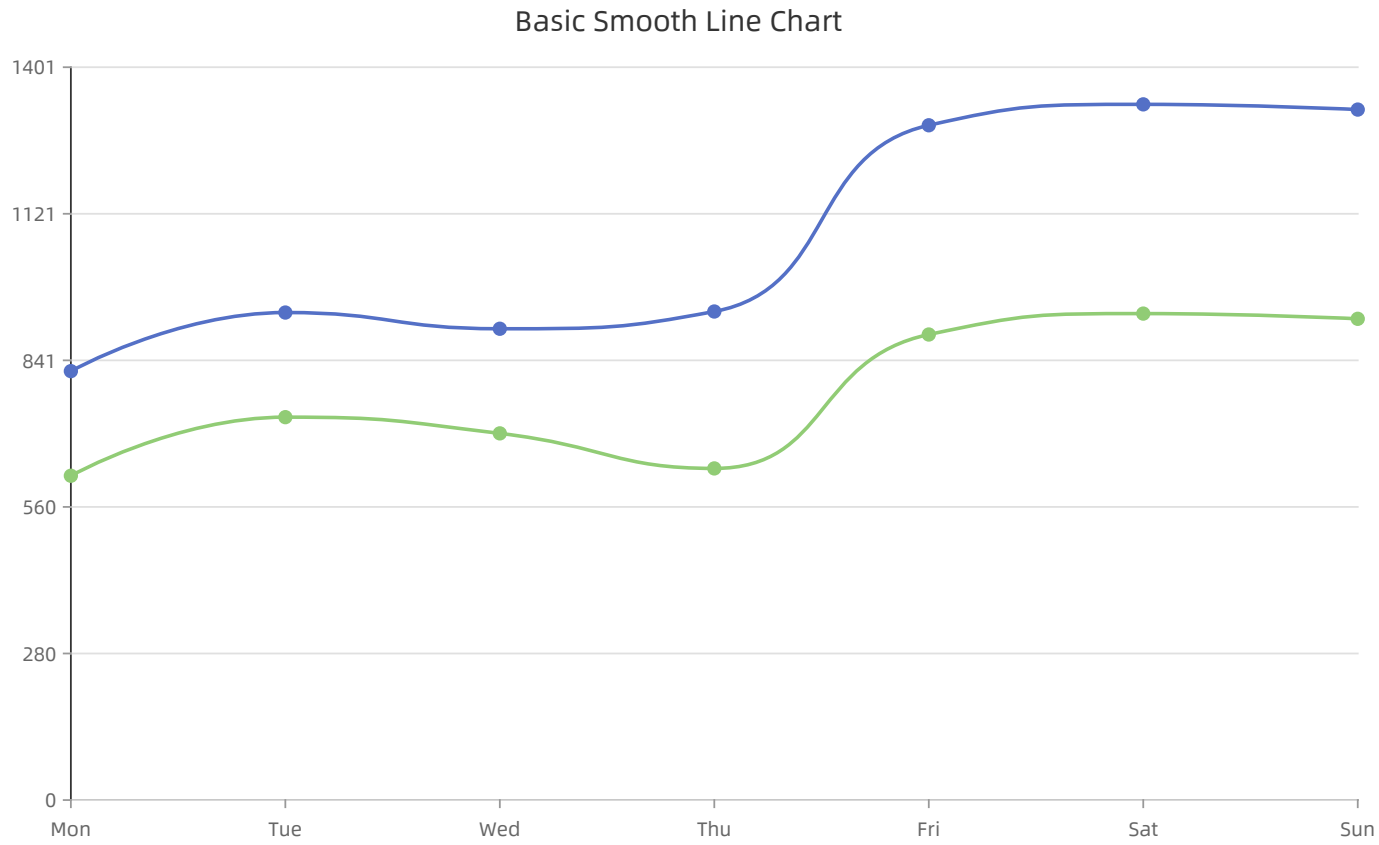
Problem	Solution
TOC page numbers are incorrect	TOC page numbers are estimated; complex layouts may cause slight deviations
Some CSS styles do not apply	webpdf supports only a subset of common CSS properties; check the supported list in the tutorial
Generated PDF file is too large	Use compressed images and avoid large raster images where possible

Chapter 10 Chart Display

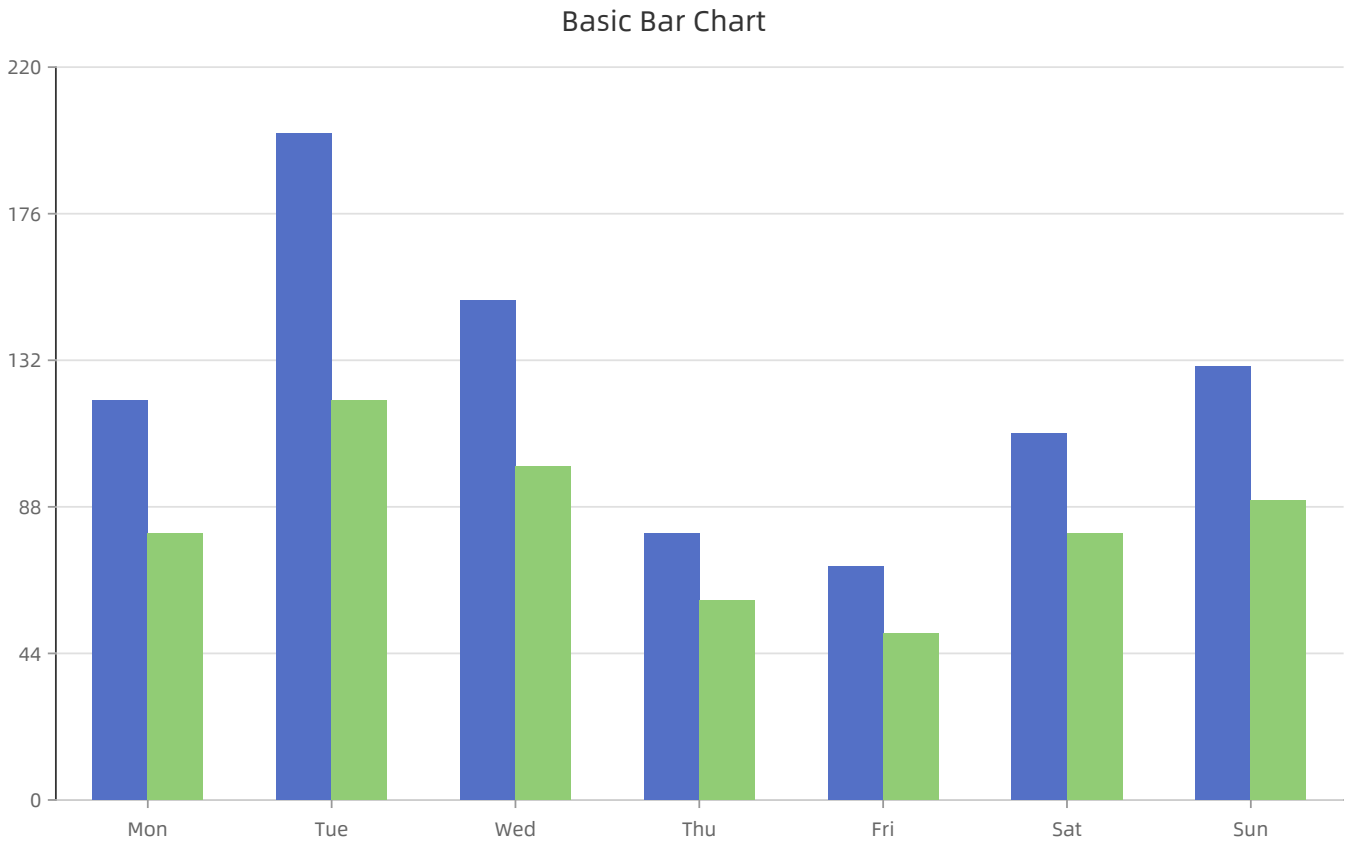
10.1 Basic Line Chart



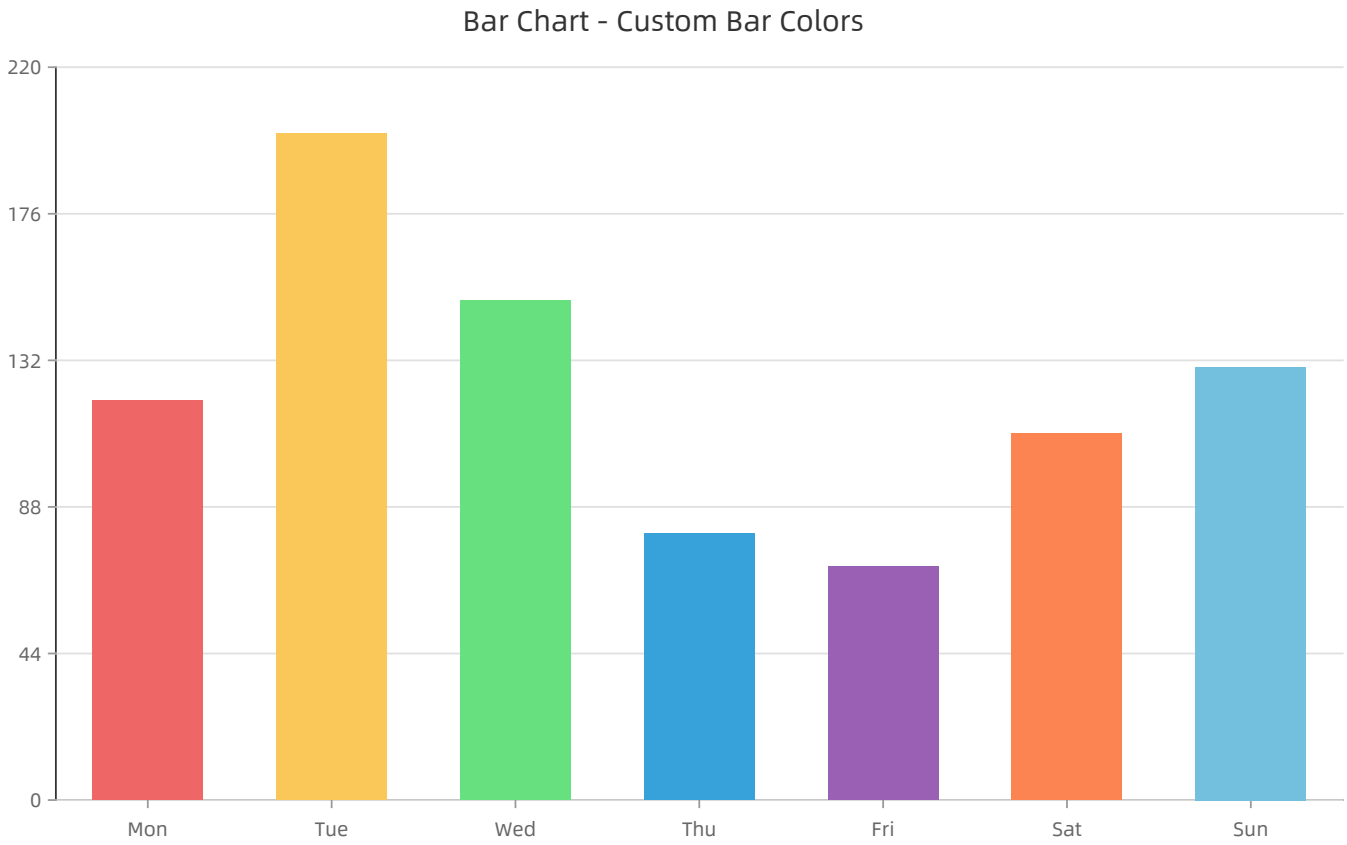
10.2 Basic Smooth Line Chart



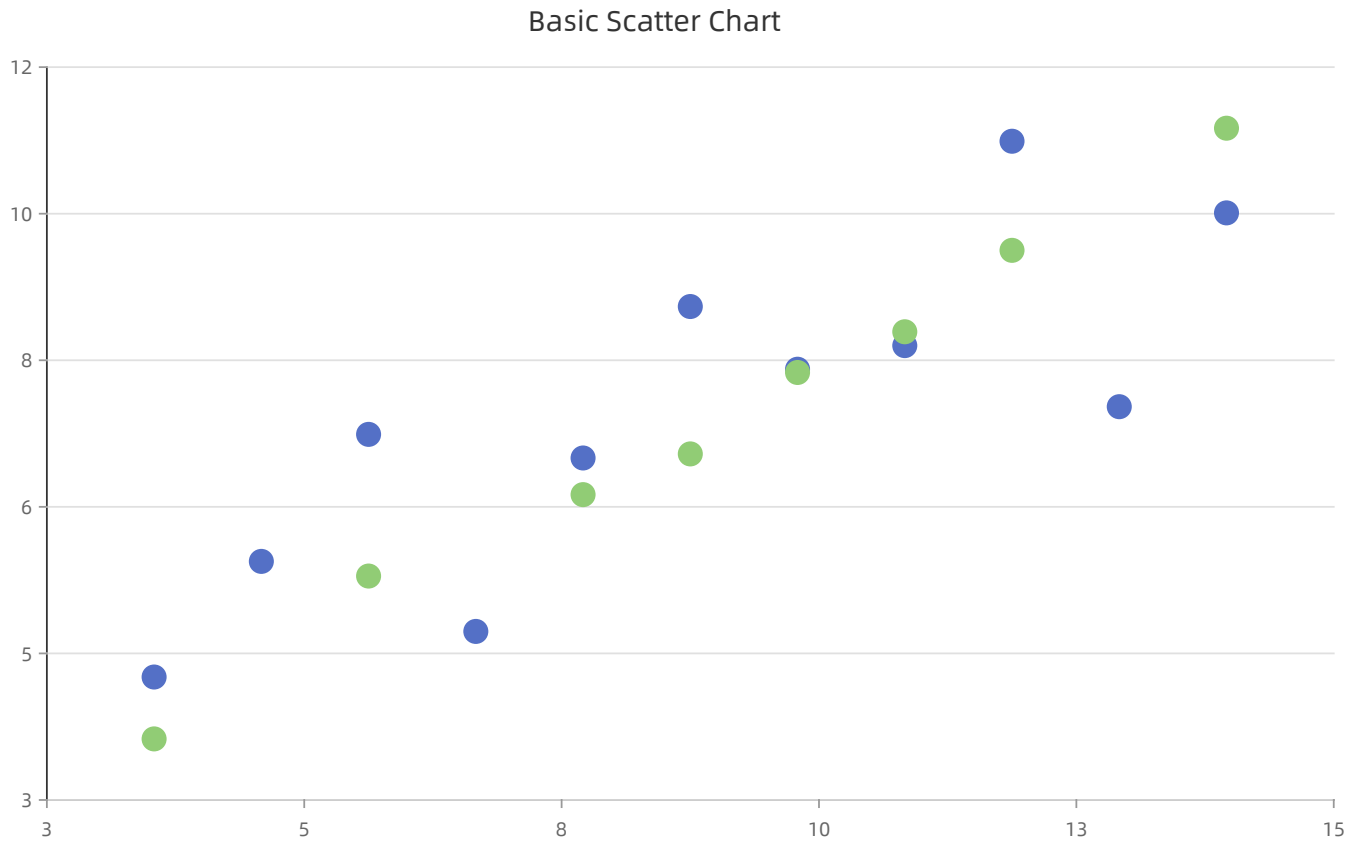
10.3 Basic Bar Chart



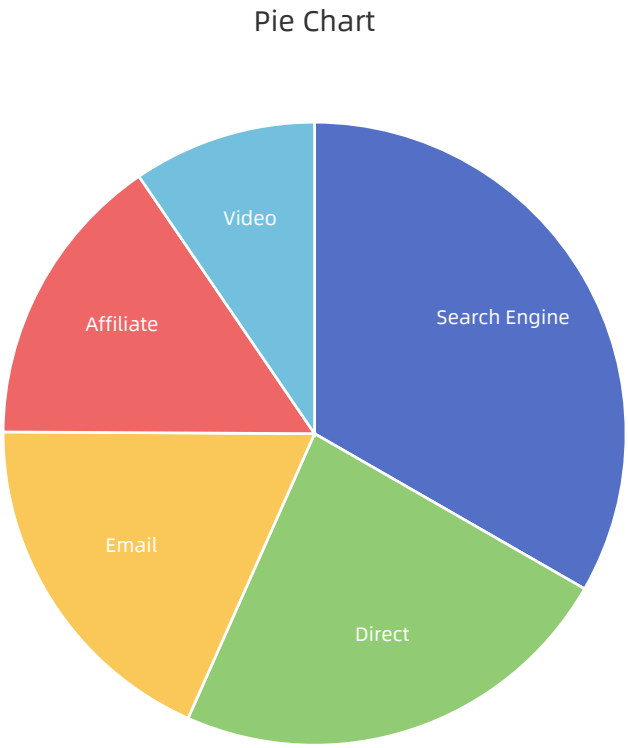
10.4 Bar Chart – Custom Individual Bar Colors



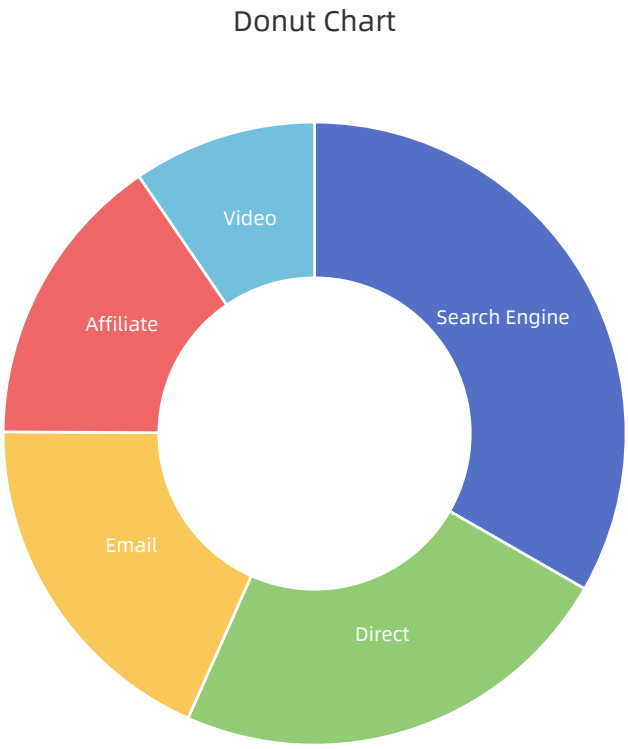
10.5 Basic Scatter Chart



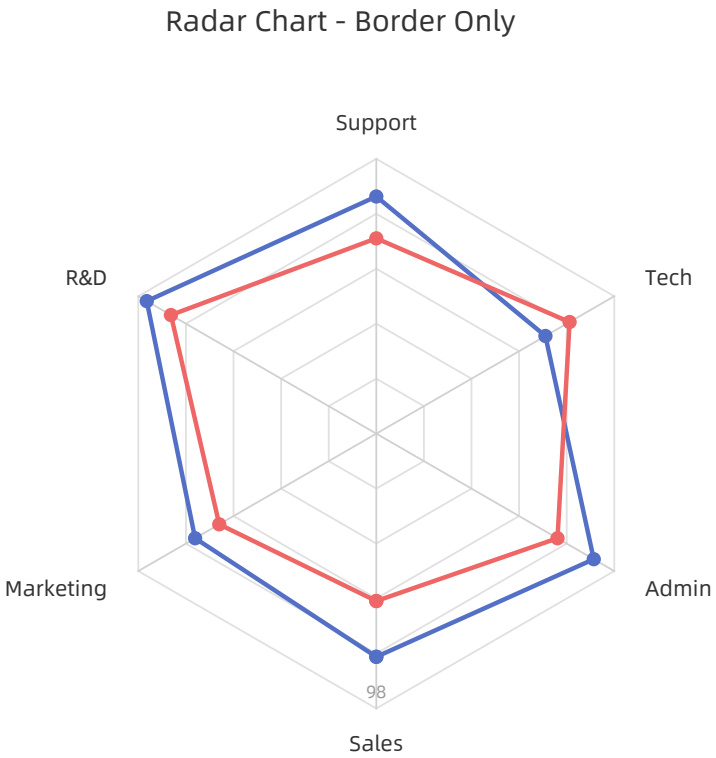
10.6 Pie Chart



10.6b Donut Chart

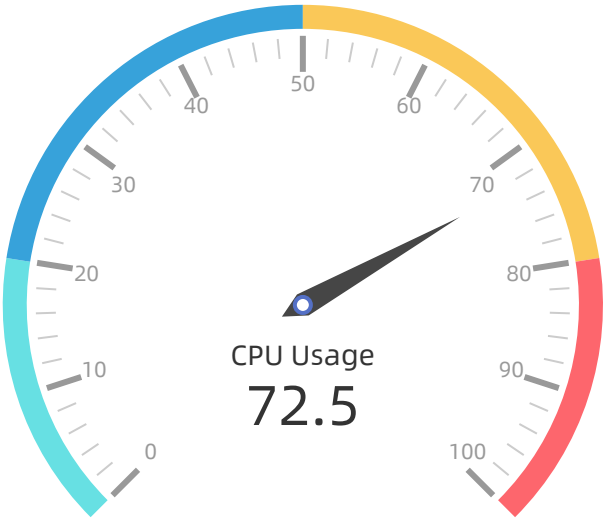


10.7 Basic Radar Chart



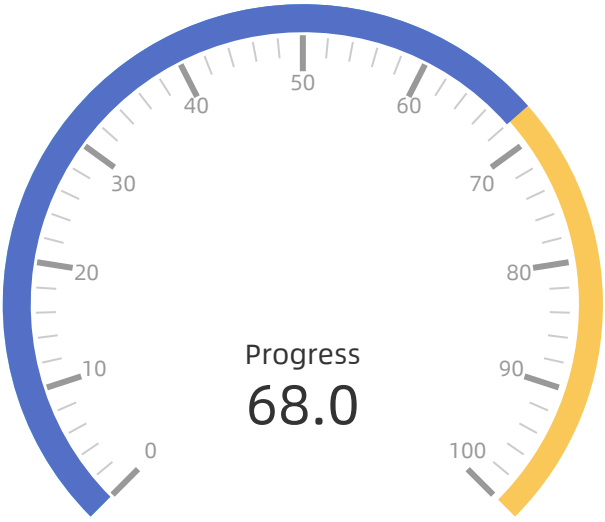
10.8 Basic Gauge Chart

Basic Gauge

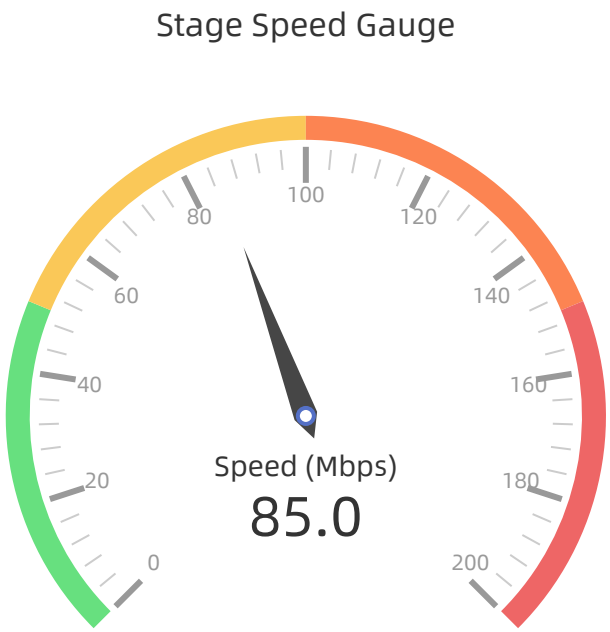


10.9 Progress Gauge Chart

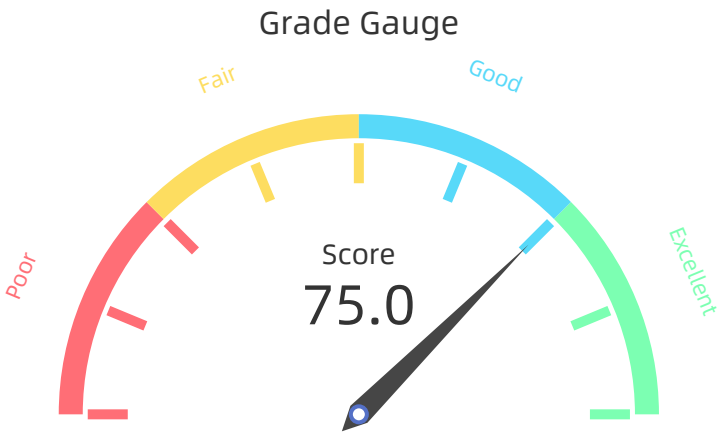
Progress Gauge



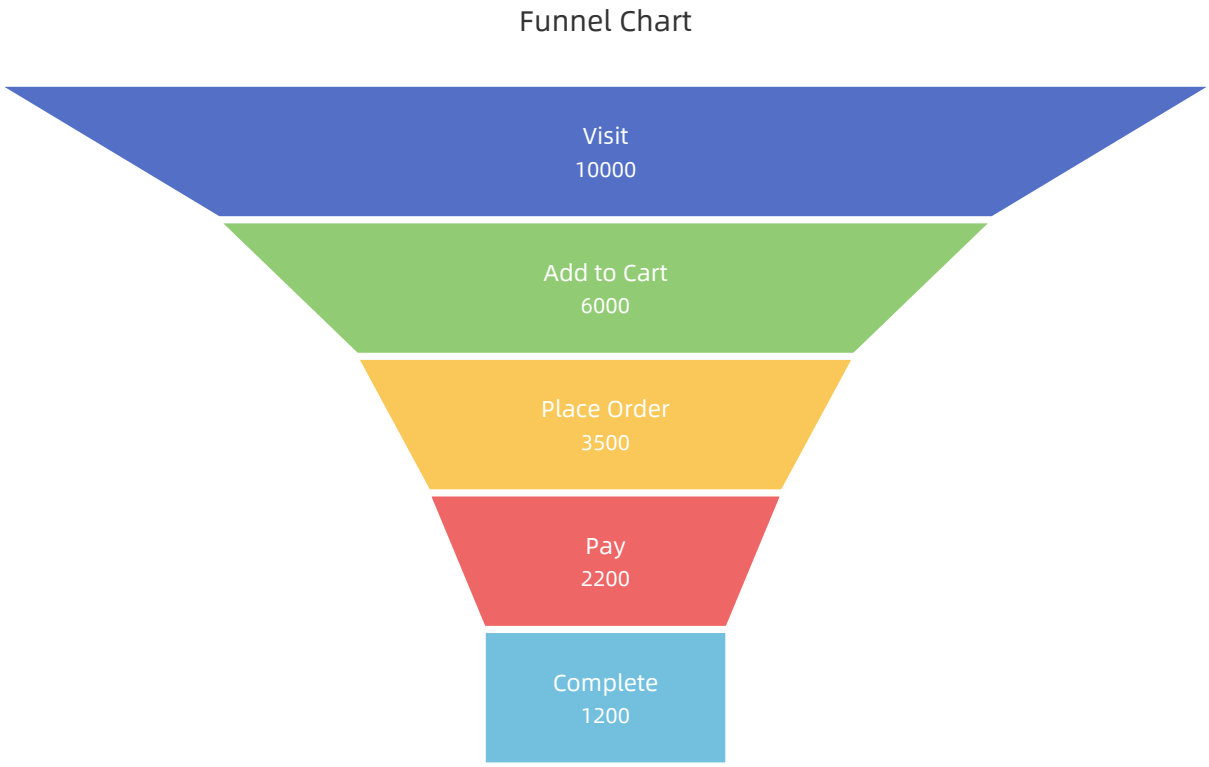
10.10 Stage Speed Gauge Chart



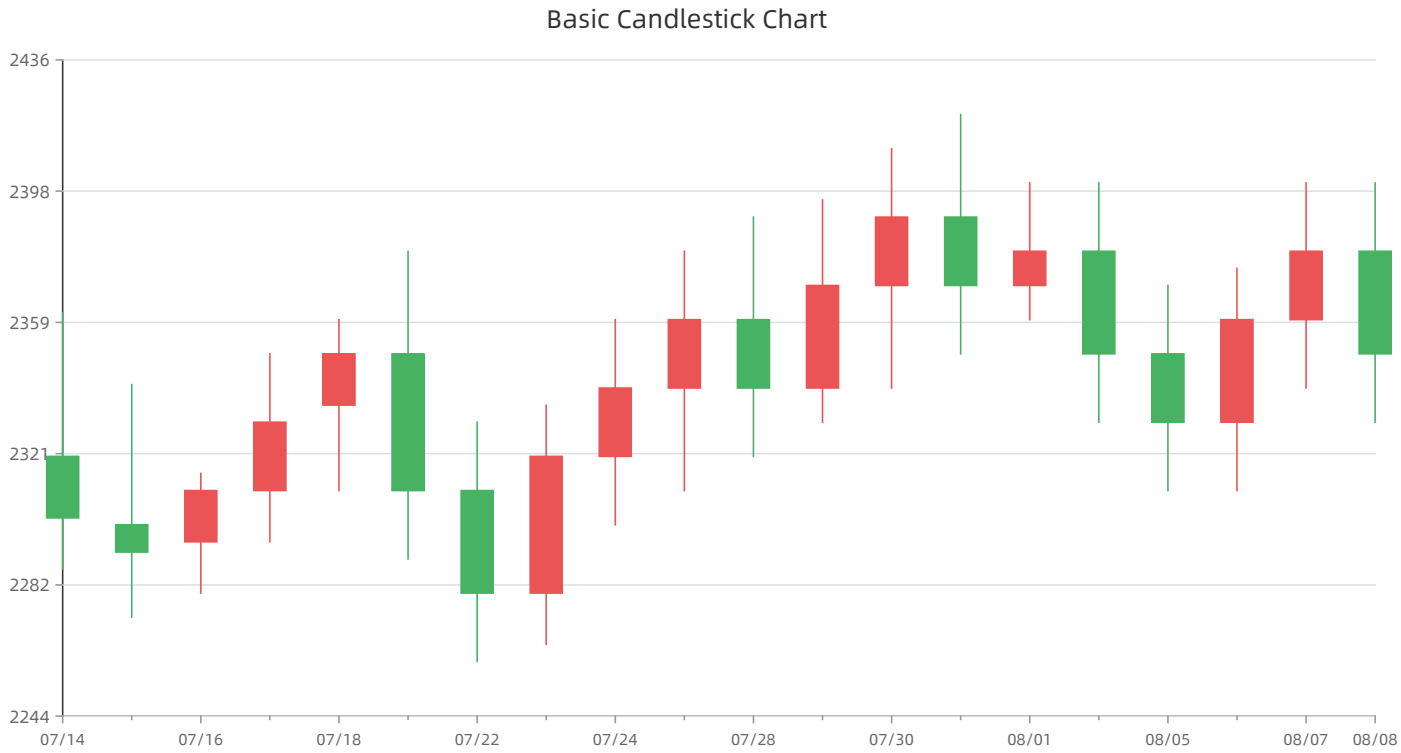
10.11 Level Gauge Chart



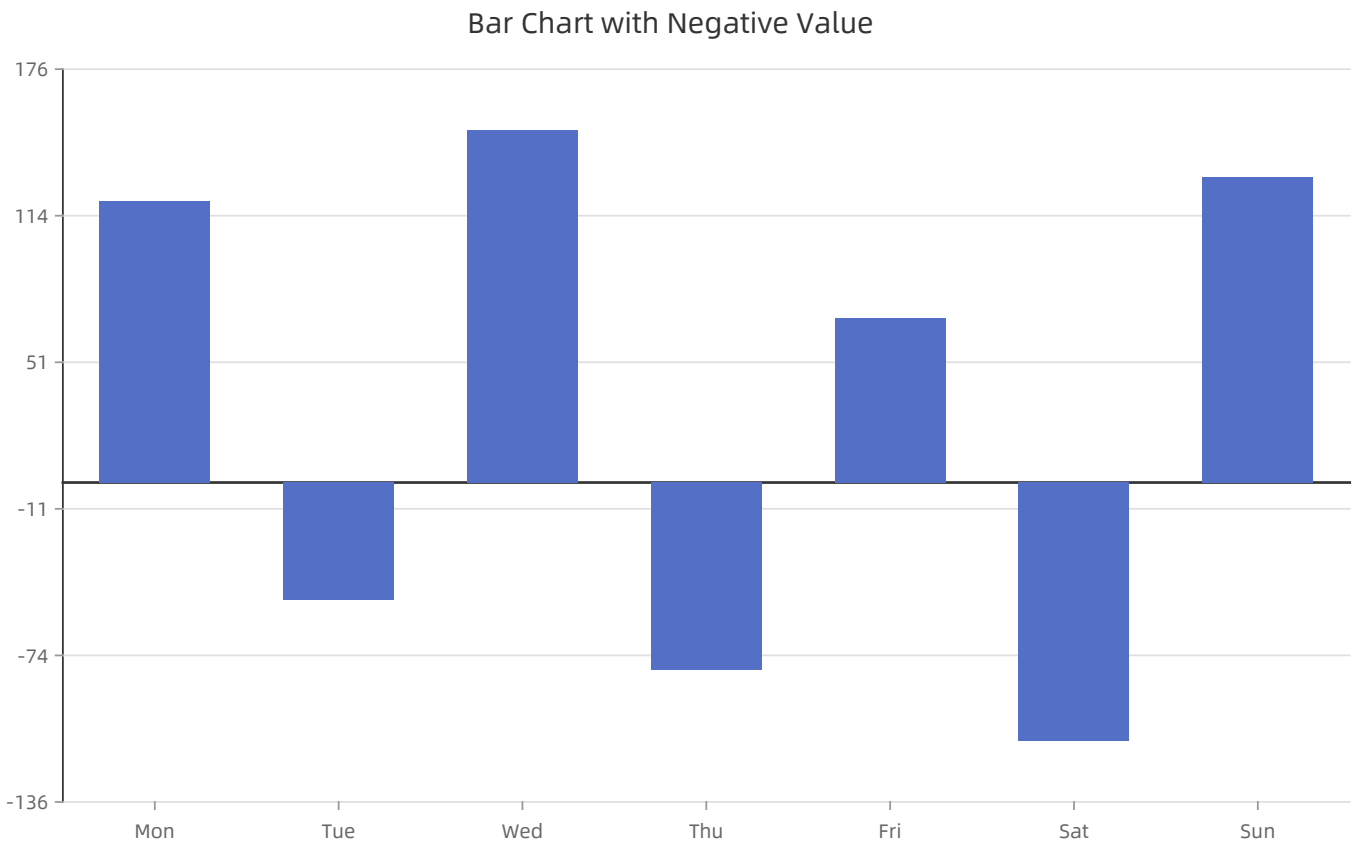
10.12 Funnel Chart



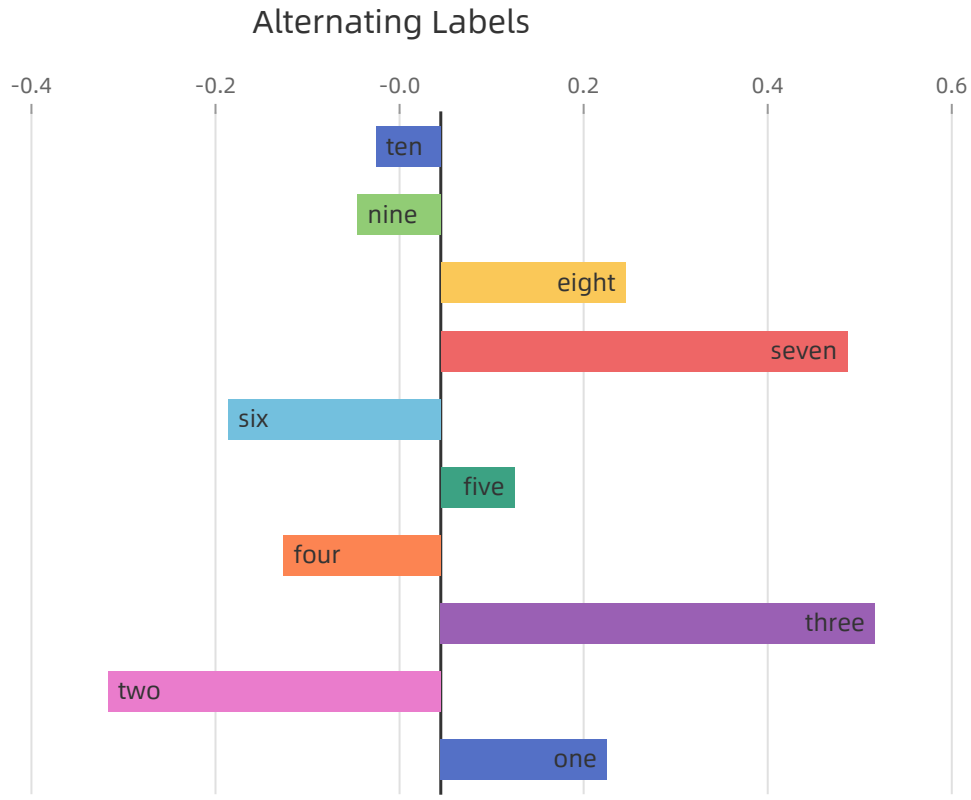
10.13 Basic Candlestick Chart



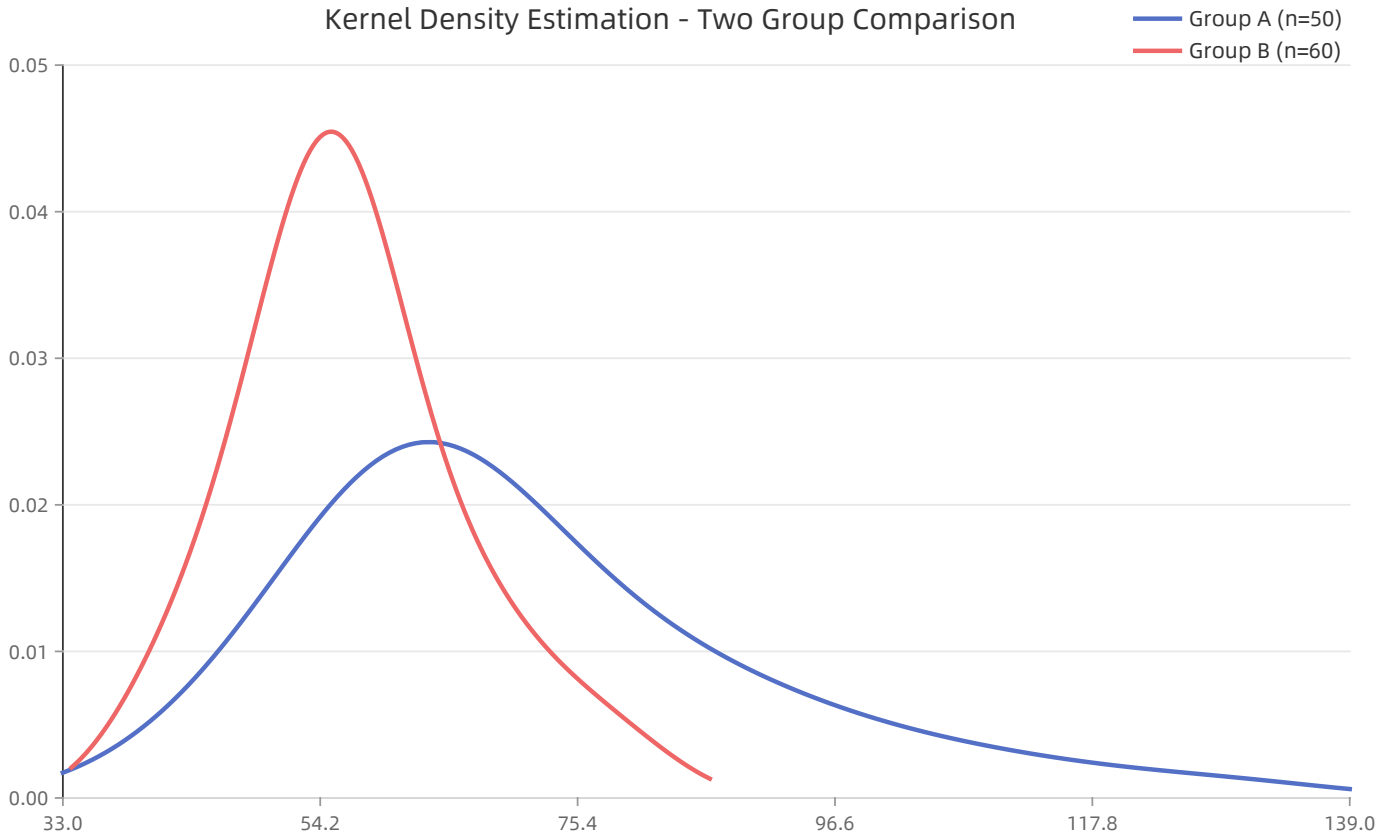
10.14 Bar Chart with Negative Values



10.15 Bar Chart with Alternating Axis Labels

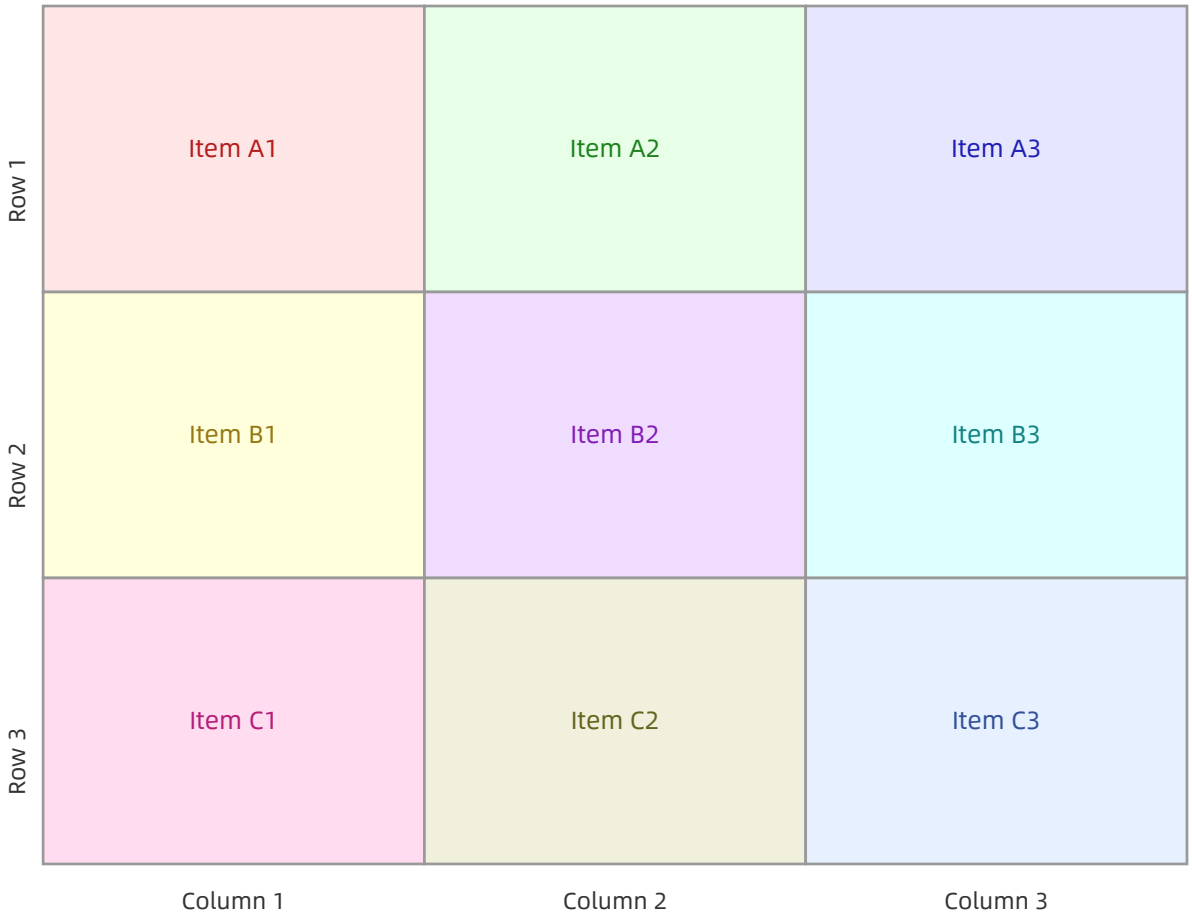


10.16 Normal Distribution (Kernel Density Estimation)



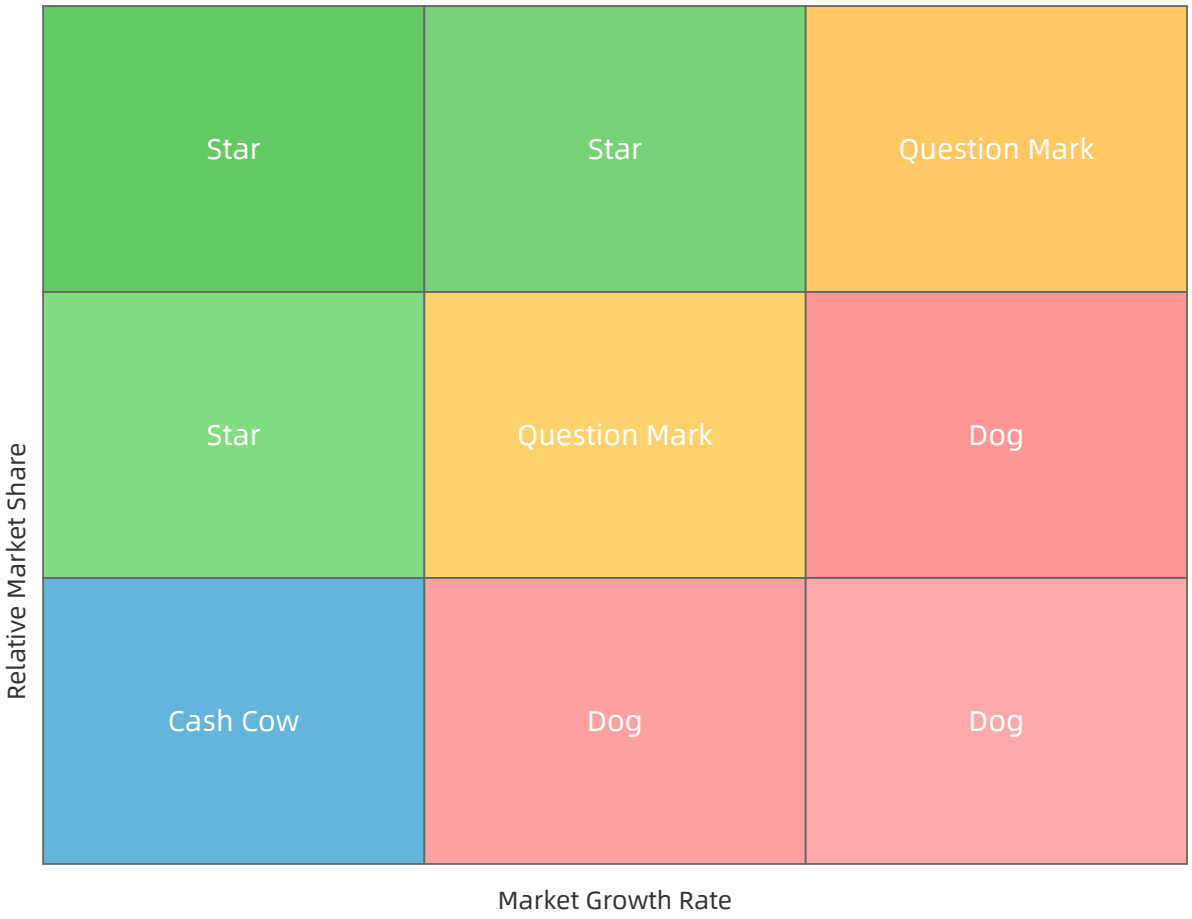
10.17 3x3 Grid Chart

3x3 Grid Chart - Custom Colors and Text

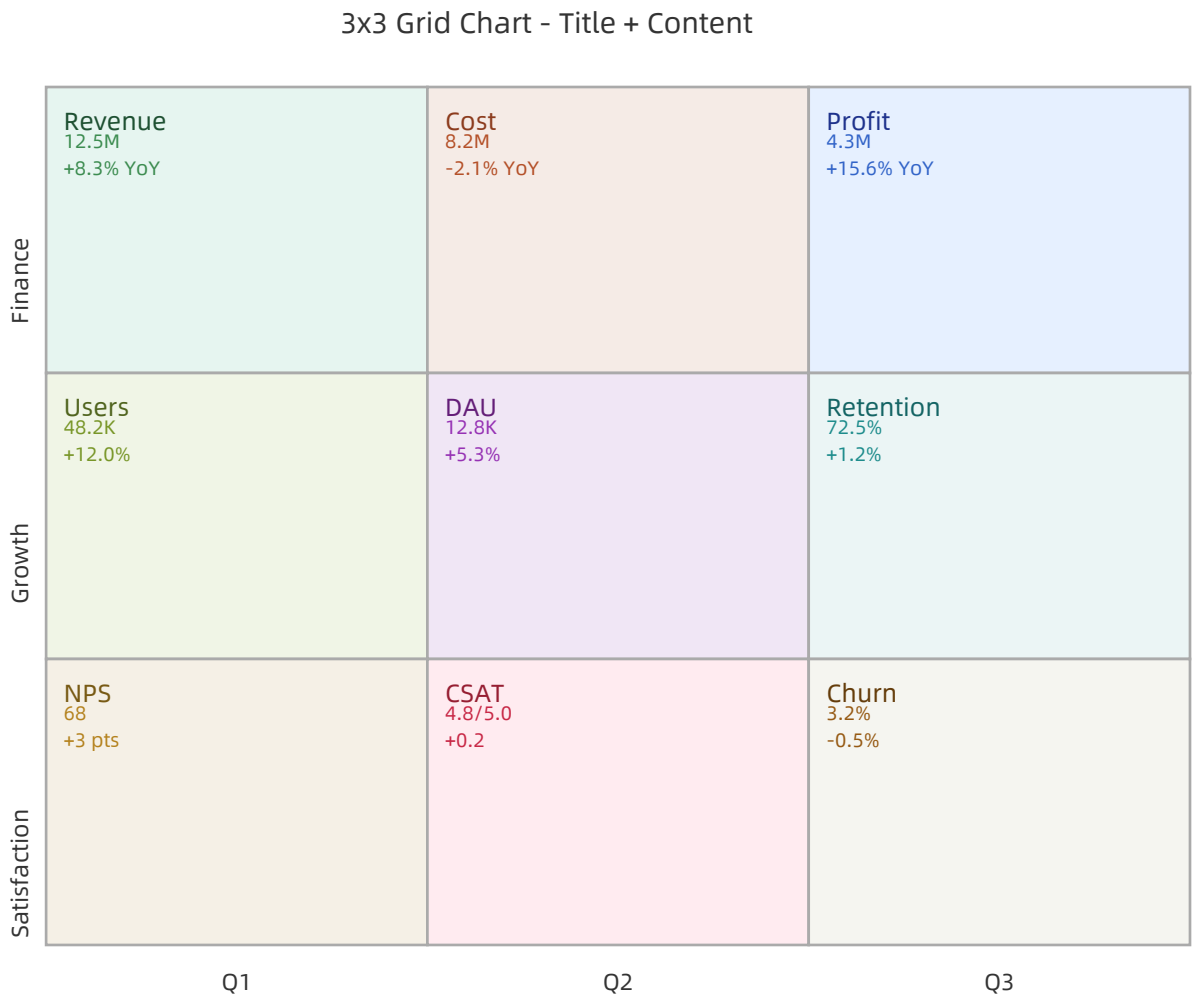


10.18 3x3 Grid Chart - Single-Label Axis Mode

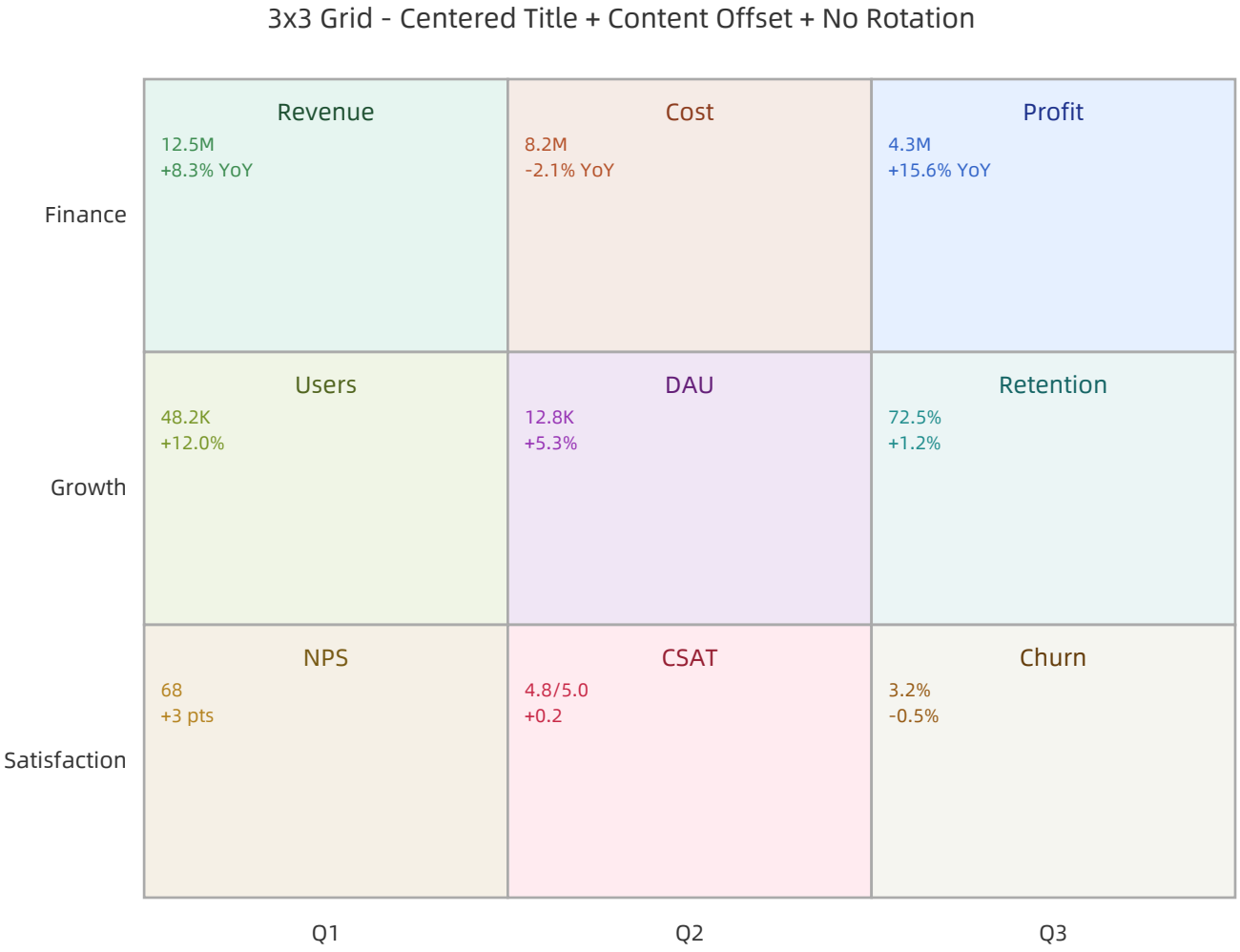
3x3 Grid Chart - Single Axis Labels



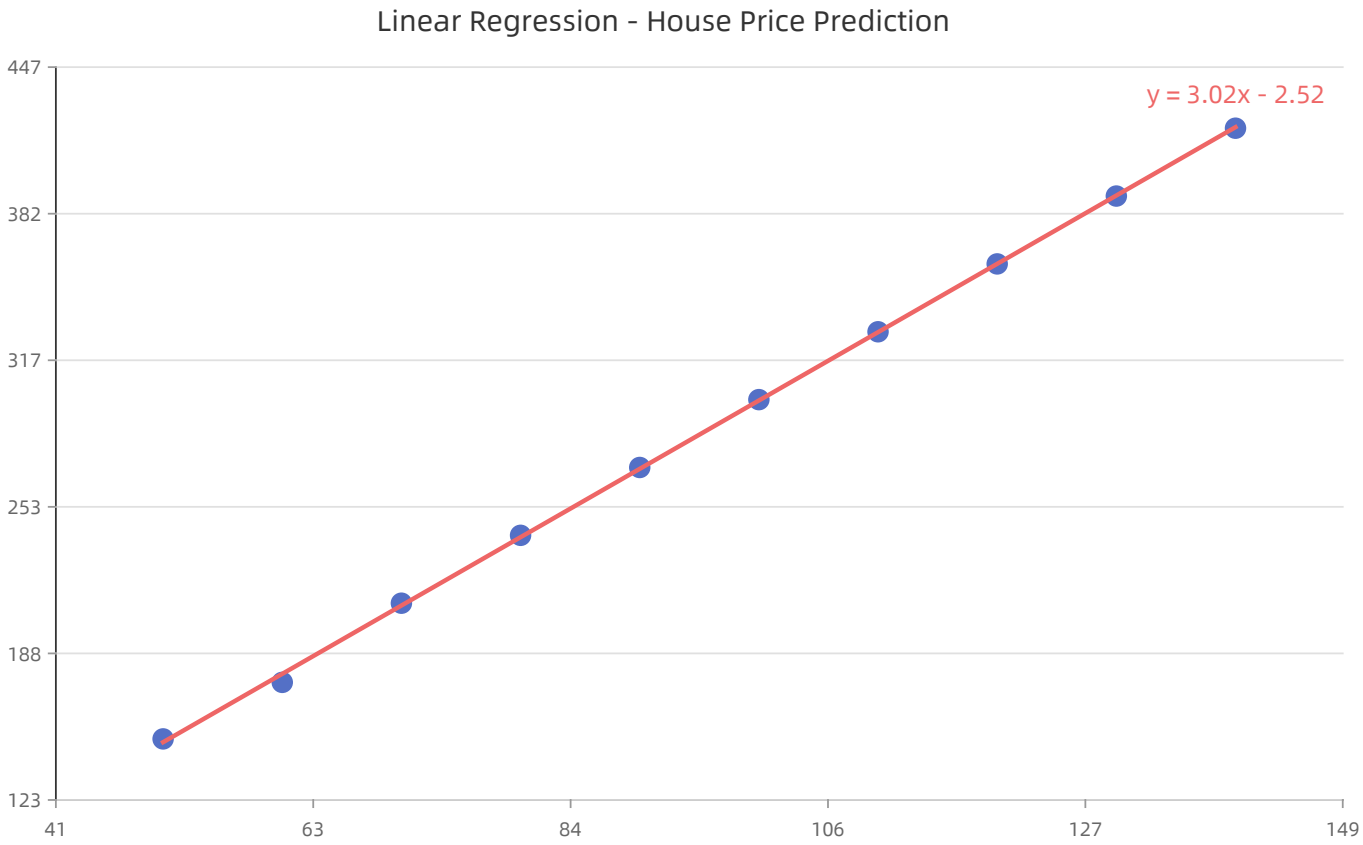
10.19 3x3 Grid Chart - Title + Content Mode (Top Aligned)



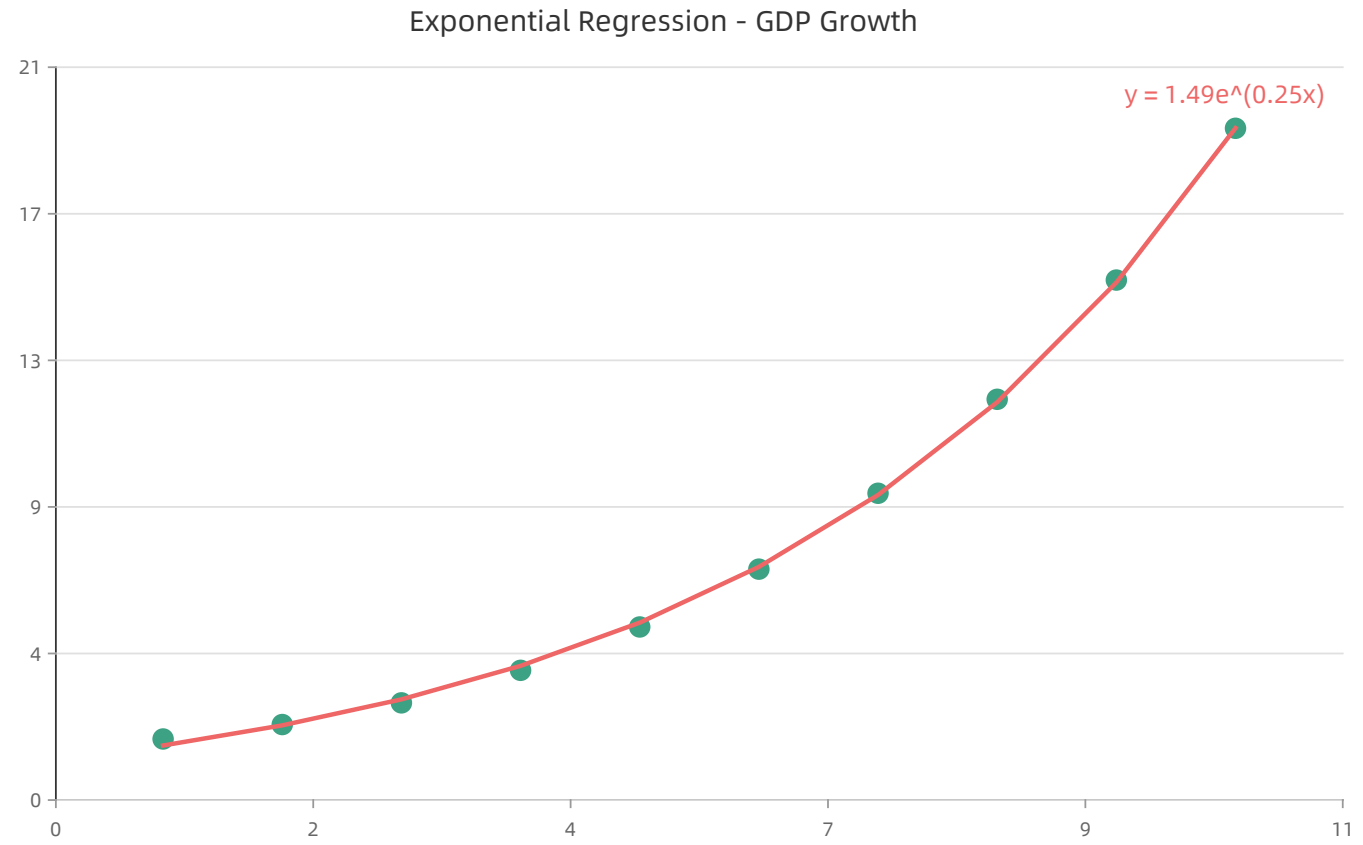
10.20 3x3 Grid Chart - Centered Title + Offset Content + Non-Rotated Y-Axis



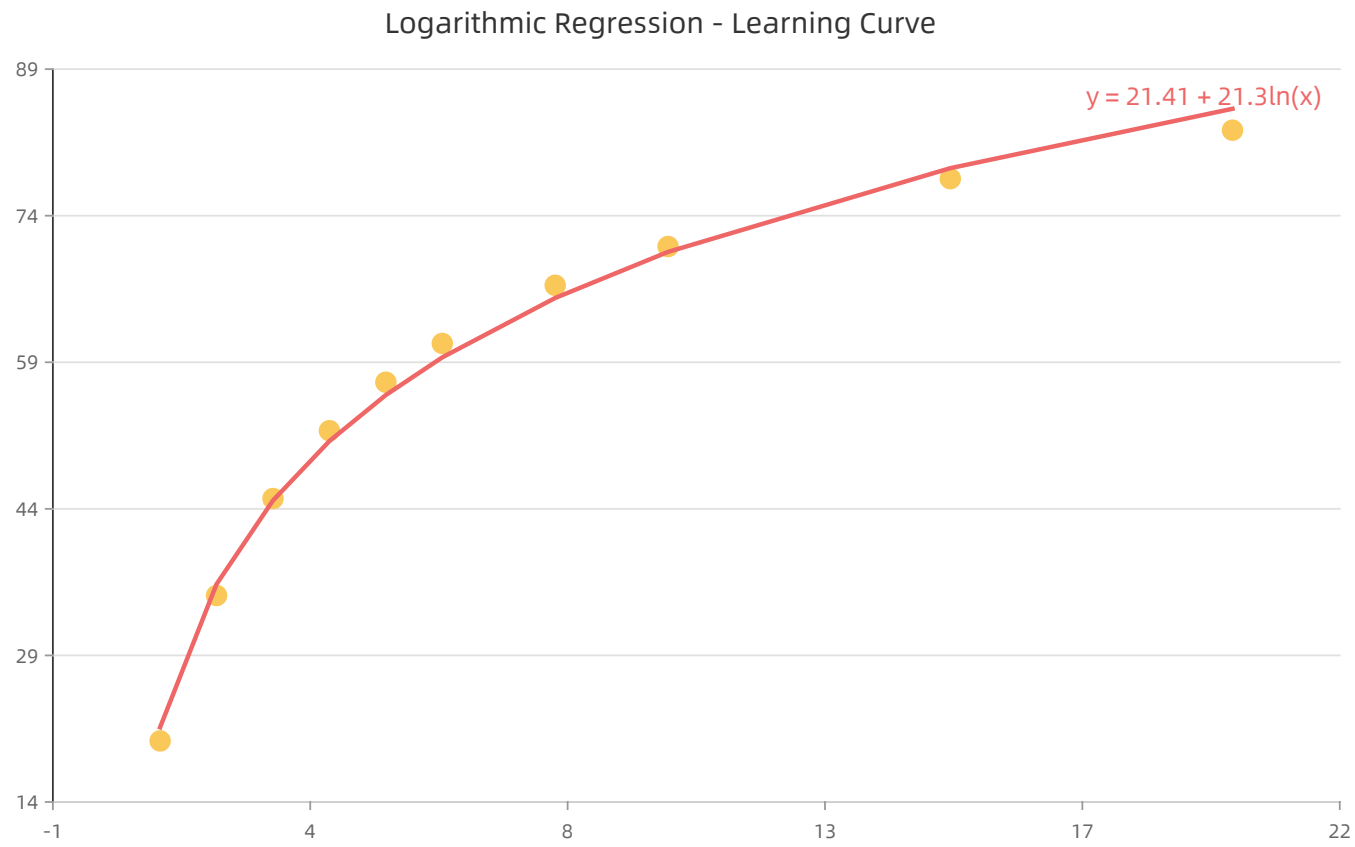
10.21 Linear Regression Plot - House Price Prediction



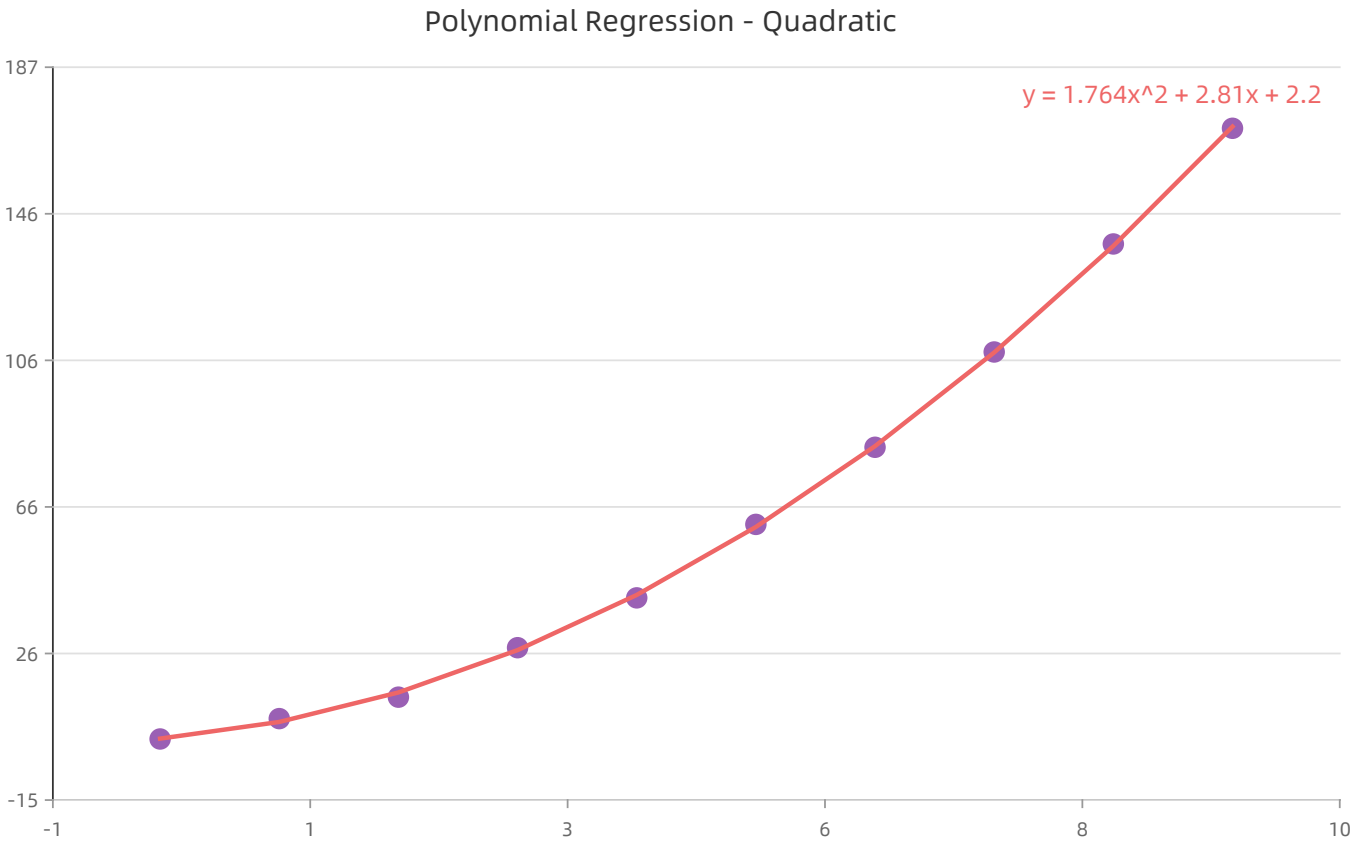
10.22 Exponential Regression Plot - GDP Growth



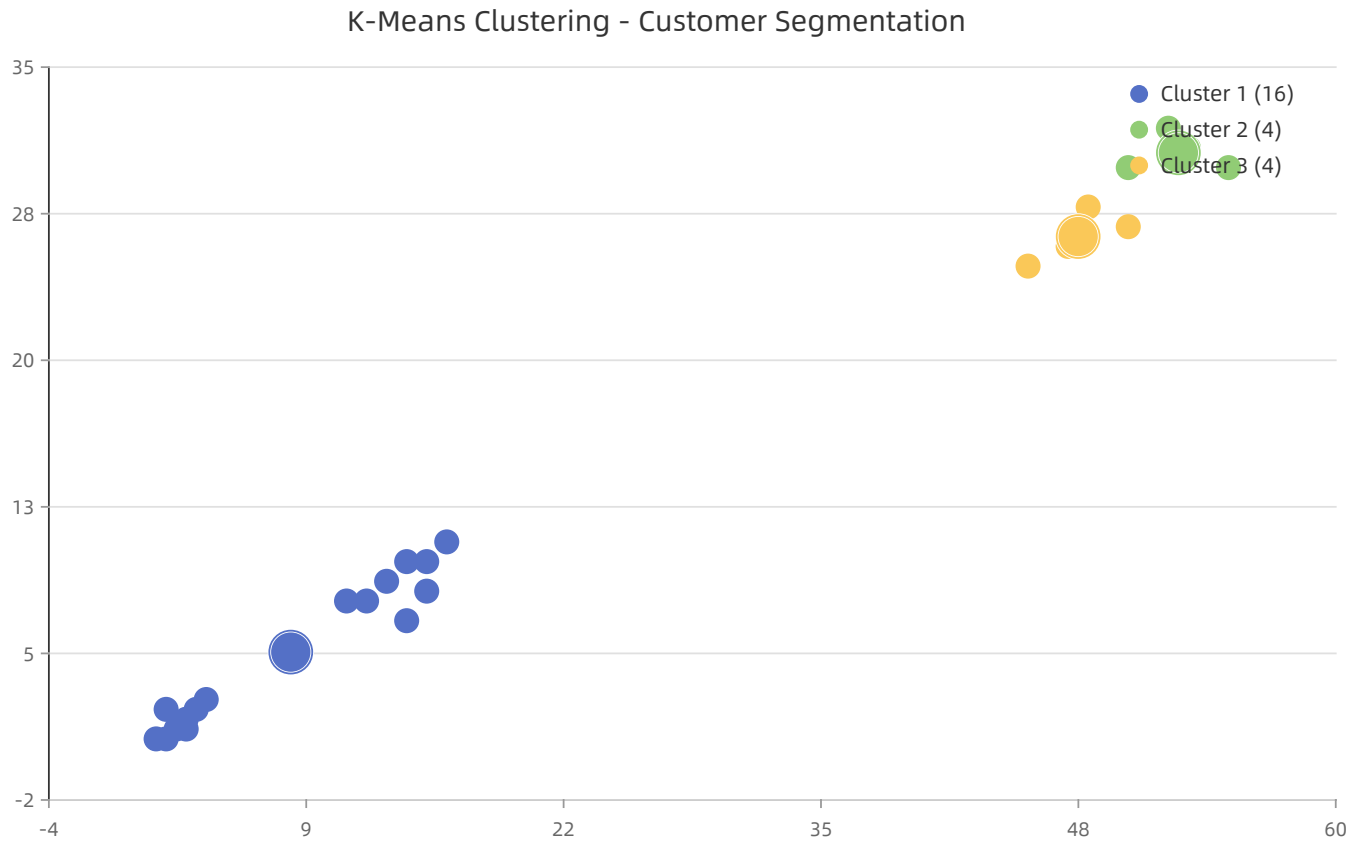
10.23 Logarithmic Regression Plot - Learning Curve



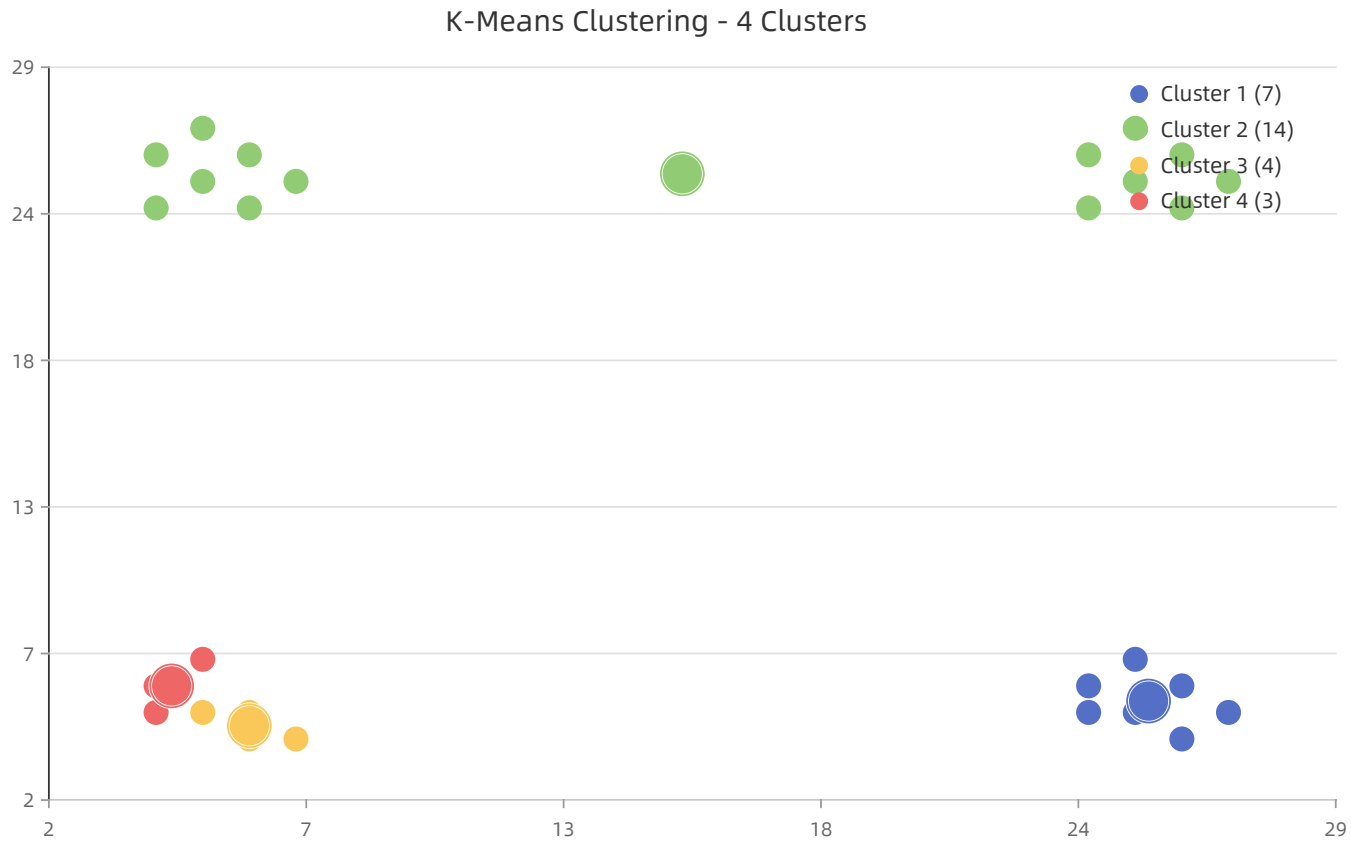
10.24 Polynomial Regression Plot - Quadratic Curve



10.25 Clustering Scatter Plot - Customer Segmentation (3 Clusters)

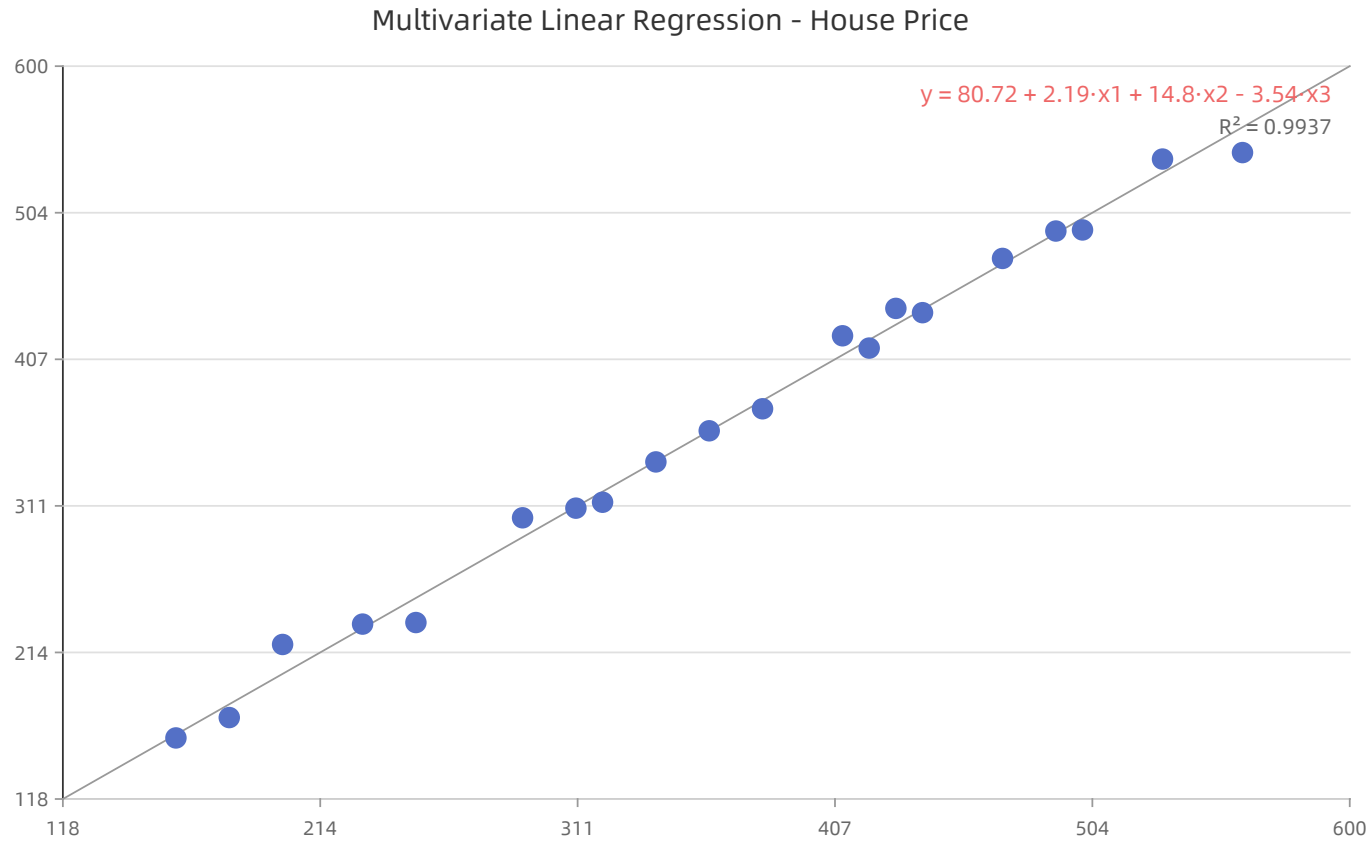


10.26 Clustering Scatter Plot - 4 Clusters



10.27 Multiple Linear Regression - House Price Prediction (Area + Bedrooms

+ Age)



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